

STARE: Semantic Augmented Reality Decision Support in Smart Environments

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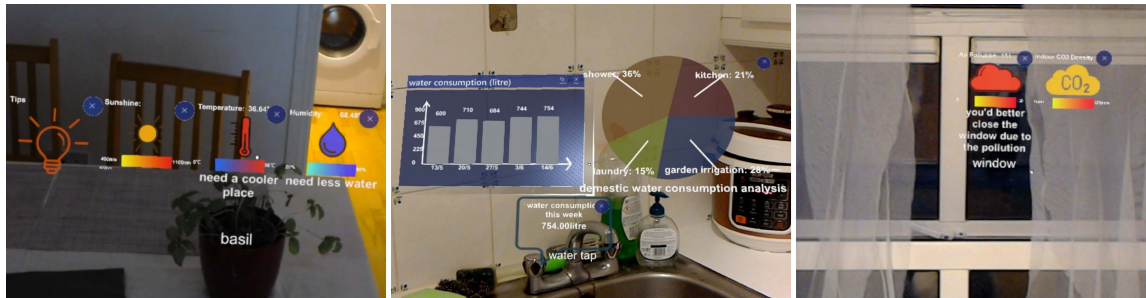


Figure 1: Augment Objects with Relevant IoT Data and Corresponding Brief Suggestions For Instant Explainable Decision Support

ABSTRACT

The Internet of Things facilitates real-time decision support within smart environments. Augmented Reality allows for the ubiquitous visualization of IoT-derived data, and AR visualization will simultaneously permit the *cognitive and visual binding* of information to the physical object(s) to which they pertain. Essential questions exist about efficiently filtering, prioritizing, determining relevance, and adjudicating individual information needs in real-time decision-making. Therefore, this paper proposes a novel AR decision support framework (*STARE*) to support immediate decisions within a smart environment by augmenting the user's focal objects with assemblies of semantically relevant IoT data and corresponding suggestions.

Index Terms: Human-centered computing—Visualization—Visualization application domains—Information visualization;

1 INTRODUCTION

The Internet of Things (IoT) provides unprecedented opportunities for the access to and conflation of a myriad of heterogeneous data to support real-time decision-making within smart environments. However, the increasing dataset scales brought by the smart environment revolution reveal the limitations of traditional 2D-screen-based smart environment data visualization interfaces: decision-makers have to spend an additional cognitive load on searching and filtering useful IoT data from the centralized long data lists.

The emergence of Augmented Reality (AR) techniques has brought in new potential decision-support solutions to spatially scatter these IoT data over a smart environment with the proximity to the data sources, such as Situated Visualization [4]. However, such scattered visualization of IoT data may sometimes lead to the separation between relevant data and the user's focus, thus generating

inconsistencies. Also, possible redundancies can gradually accumulate when the decision-maker switches attention from their currently focused context to discover the relevant information among other irrelevant IoT data scattered over the smart environment. To avoid these possible redundant and inconsistent artifacts which contribute to information overload factors and hindered decision-making, we propose the *STARE* (SemanTic Augmented REality) decision support framework to augment the user's focal context with ubiquitous and seamless decision support.

Decision support systems (DSS) are commonly defined as computer-based systems used for assisting decision making [2]. Prior data-oriented AR smart environment DSS demonstrated that they could assist in decision-making by providing data retrieval and information analysis [1] but left the vital decisions to expert users. However, non-expert users may benefit more from the higher-level model-oriented decision support such as consequence simulation and suggestions [1] for non-vital decisions. We found no prior AR smart environment interfaces provided such high-level decision support by reviewing the current literature. Therefore, after establishing semantic object-data associations for the focus augmented IoT data, *STARE* further encapsulates an ontology-based suggestion model to provide explainable suggestions targeted for non-expert users' focal decision contexts within a dynamically changing smart environment.

2 SYSTEM OVERVIEW

By implementing *STARE* on a Microsoft HoloLens, this system encapsulates the focus augmentation and a novel explainable suggestion model to shorten the physical and semantic distance between the user's focal object and its relevant decision support data. To achieve the focus augmentation strategy, *STARE* applies *focus + voice command* modality to allow for the hands-free triggering of the AR information over users' focal objects. This system consists of five main modules (Fig. 2): object classifier, semantic annotator based on a lightweight ontology, decision support component, Bluetooth Low Energy (BLE) sensor data scanner, and AR information visualization module. This system allows the users to freely explore the smart environment by invoking the decision-support information for the objects they are currently focusing on. For example, a user wants to open the window but is not sure about outdoor air

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quality, so they use a voice command to trigger a decision-support AR annotation about the window, which superimposes indoor and outdoor air quality data over the windows and shows a suggestion to close the window due to outdoor air pollution (Fig. 1).

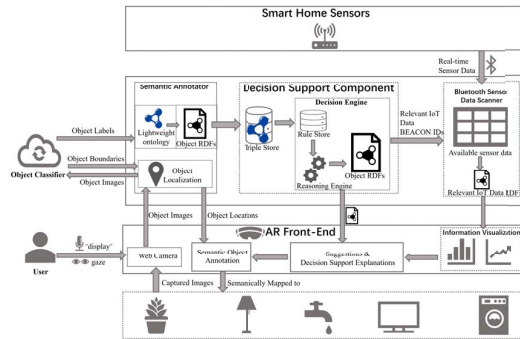


Figure 2: System Architecture Diagram

2.1 Object Classifier

After detecting the voice command, the web camera captures an image for the current field of view and then sends it to the Microsoft Azure Custom Vision¹ cloud service for object classification. To avoid information overload caused by excessive AR superimposition over all recognized objects, *STARE* defines one object list for each given smart home context to filter decision-involved objects. Moreover, one matching percentage threshold and object depth threshold are also defined to filter out the unclear objects. For each matching object recognized from this image, one bounding box is returned by the object classifier, which is then used for the object localization in the physical world by combining the depth data.

2.2 Lightweight Ontology & Semantic Annotator

The proposed lightweight ontology aims at providing a semantic structure for the data collected from the different sensors and serves as a stepping stone towards the more elaborated reasoning rules of the Decision Support component of *STARE*. The semantic annotator extends the Semantic Sensor Network (SSN) Ontology to formally express the events captured by the sensors, while extending the SOSA Ontology (Sensor, Observation, Sample, and Actuator) [3] to formally express and record the object recognition and localization events captured by the object classifier. Simultaneously, the *STARE* lightweight ontology transfers the object label into Resource Description Framework (RDF) documents (Fig. 2) to ensure that the decision support component can easily consume them. The *STARE* lightweight ontology further supports observations registered by the sensors and provides a semantic structure of the sensors by combining the vocabularies from SSN and SOSA.

2.3 Decision Support Component

The decision support component provides a novel suggestion model to achieve high-level decision support within a dynamically changing environment. As Fig. 2 shows, the decision support component incorporates a triplestore and a decision engine to generate suggestions and explanations. The object RDF documents generated by the semantic annotator are stored in this triplestore, which serves as the repository to feed the decision engine. This decision engine comprises a rule store and a reasoning engine to construct the multi-dimensional semantic associations between the object and its relevant IoT data. The rule store assigns decision rules to the

object RDF, based on which the reasoning engine makes logical inferences. According to the decision rules, the reasoning engine annotates this object RDF document with a list of relevant IoT data and their descriptors. This relevant IoT data list contains environmental parameters that may change the object's states (e.g., sunshine and moisture for indoor plants; fridge temperature for the fridge) or interfere with the usage decision about this object (e.g., air quality for air purifier and windows; energy consumption for the home appliances). For each annotated IoT data, the data descriptor contains the objective properties, such as the sensor/device name and a BEACON ID used for BLE data scanning (Fig. 2).

2.4 AR Explainable Decision Support Data Visualization

Utilizing the decision rules and IoT data descriptors derived from the decision engine, the AR front-end can superimpose suggestions over the focal object to support decisions about it (Fig. 1). However, such high-level suggestions or recommendations without explanation do not generate trust from a decision-maker. Accordingly, *STARE* explains the provided brief suggestions with an assembly of the relevant IoT data defined in the object RDF, which work as the decisive input values that may affect the resulting advice [5]. Among these decisive input values, the key features that determined the decisions are then highlighted with high color codes [6] and alerts (Fig. 1) to clearly explain the decision logic behind the suggestions.

To retain data details without causing visual clutter, this system provides a simplification for all superimposed data and allows for further investigation in a details-on-demand manner. As Figure 1 shows, by clicking (gesture interaction) on any IoT data icon, detailed information will be visualized in line charts, histograms, or pie charts, which are determined by the data types.

3 CONCLUSIONS

This paper presented an innovative AR smart environment decision support framework (*STARE*) which incorporates a novel suggestion model and a focus augmentation modality to allow for immediate-continuous decision support within the user's currently investigated context. Further enhancements to *STARE* incorporating machine learning will enable the automatic construction of semantic data-object associations. This enhancement will ensure *STARE* extensibility that allows smart environments to deliver *truly* intuitive interfaces for a future empowered ubiquitous IoT world.

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¹<https://azure.microsoft.com/services/cognitive-services/>